

Reinforcement Learning in Wargames and Military Simulations

A survey of methods, case studies, metrics, and outlook for DoD wargaming (2020 to 2026)

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Scope	Peer-reviewed papers, DoD program pages, and FFRDC reporting on RL applied to wargaming and military simulation, 2020 to 2026.
Method	Literature scan across IEEE, arXiv, DARPA, AFRL, DIU, and journals such as the Journal of Defense Modeling and Simulation.
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Executive Summary

Between 2020 and 2026, reinforcement learning (RL) moved from controlled academic demonstrations to operational and pre-operational use across U.S. Department of Defense (DoD) wargaming and simulation programs. The decisive academic milestone was the DARPA AlphaDogfight Trials in 2020, where a deep RL agent from Heron Systems defeated a U.S. Air Force F-16 weapons instructor five to zero in simulated within-visual-range combat¹. That result anchored a wave of follow-on work, including the DARPA Air Combat Evolution (ACE) program flying autonomy on the X-62A VISTA testbed in 2022 and 2023², and DARPA SCEPTER, which targets machine-generated courses of action explored 10,000 to 100,000 times faster than real time at scales up to 10,000 agents³.

The literature converged on a small number of dominant methods: self-play with proximal policy optimization (PPO), hierarchical RL (HRL) that mirrors military command echelons, multi-agent RL (MARL) for coordinated forces, and increasingly hybrid pipelines that pair RL or scripted agents with large language models (LLMs)^{4,5}. Reported wins include superhuman performance in air combat sims, five-times throughput gains in staff wargames at the U.S. Army Command and General Staff College using an LLM-driven wargame in late 2025⁶, and operational fielding of the U.S. Air Force WarMatrix wargame system at the GE 26 Benchmark Wargame in March 2026⁷. Persistent challenges remain: sample inefficiency, sim-to-real transfer, reward hacking, brittle behavior under adversarial perturbations⁸, and limited explainability for commander trust.

The near-term DoD outlook (2026 to 2028) is dominated by hybrid LLM plus RL planners (Thunderforge, WarMatrix, CGSC pilots), faster simulators that lift the sample-efficiency ceiling, and a hard push toward explainable, adversarially robust agents that commanders can actually use for course-of-action analysis^{9,10}.

1. DARPA AlphaDogfight, Wikipedia, https://en.wikipedia.org/wiki/DARPA_AlphaDogfight

2. DoD AI agents successfully pilot fighter jet, AFRL, afrl.af.mil/News/Article-Display/Article/3297364
3. SCEPTER program page, DARPA, darpa.mil/research/programs/scepter-strategic-chaos
4. Pope et al., Hierarchical Reinforcement Learning for Air-to-Air Combat, arXiv:2105.00990, arxiv.org/abs/2105.00990
5. Black, Mastering the Digital Art of War, arXiv:2408.13333, arxiv.org/abs/2408.13333
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7. USAF GE 26 showcases WarMatrix, Air Combat Command, acc.af.mil (Article 4150428)
8. Dabholkar et al., Adversarial Attacks on RL Agents for Command and Control, JDMS, arXiv:2405.01693, arxiv.org/abs/2405.01693
9. DIU Thunderforge announcement, Mar 2025, diu.mil (Thunderforge)
10. Lamparth et al., Human vs Machine: Language Models and Wargames, arXiv:2403.03407, arxiv.org/abs/2403.03407

1. Background and Why Reinforcement Learning

DoD wargaming exercises the human side of military judgment by exploring conflict under rules, scenarios, and adjudication. The classic critique, repeated across recent FFRDC commentary, is that human-only wargames are slow, sparsely sampled, and biased by the personalities in the room. Most current campaign analyses run a single scenario across a few weeks. Reinforcement learning offers three properties that map cleanly onto that problem¹¹:

- Self-improvement through interaction. RL agents learn directly from simulated combat outcomes, without requiring labeled expert traces. This sidesteps the expert-data bottleneck that constrains supervised approaches in classified domains.
- Scale. Once trained, an RL agent can replay millions of scenarios in hours, exposing fragile assumptions in plans and revealing novel courses of action that human planners would not enumerate.
- Adaptive adversaries. Self-play and population-based training generate red teams whose tactics evolve, replacing brittle scripted opponents that exercise only the behaviors the developer thought of.

The 2018 DoD AI Strategy and the 2023 DoD Data, Analytics, and AI Adoption Strategy explicitly named wargaming and decision support as priority insertion points, which catalyzed the program lineage covered in this report¹².

2. Methods Taxonomy

Recent work clusters into five algorithm families. Most fielded systems combine two or more.

2.1 Single-Agent Deep RL (PPO, SAC, DQN variants)

Proximal policy optimization (PPO) is the workhorse for continuous control problems such as dogfighting, missile guidance, and UAV maneuvering. Soft actor-critic (SAC) and maximum-entropy variants are common where exploration of unusual tactics matters. Heron Systems used deep RL on a single F-16 control policy to win AlphaDogfight¹. Lockheed Martin's second-place entry layered maximum-entropy RL with reward shaping from subject matter experts⁴.

2.2 Hierarchical Reinforcement Learning (HRL)

HRL decomposes the policy into a high-level controller selecting goals or sub-policies and low-level controllers that execute primitives. This maps naturally onto military command echelons. Pope et al. (Lockheed Martin, 2021) used HRL to integrate modular low-level policies for air-to-air combat⁴. Black (Naval Postgraduate School, 2024) extended HRL with dimension-invariant observation abstractions to scale combat simulation agents from squad to operational level^{5,13}.

2.3 Multi-Agent Reinforcement Learning (MARL)

MARL targets coordinated forces. Centralized training with decentralized execution (CTDE) is the dominant pattern, often using MAPPO, QMIX, or COMA. Reported uses include cooperative naval combat with attention-based credit assignment¹⁴, ground counterforce allocation, and coalition cyber defense with hierarchical MARL¹⁵.

2.4 Self-Play and Population-Based Training

Inspired by AlphaStar and AlphaZero, self-play and league play generate non-stationary adversaries that drive strategic novelty. Fictitious self-play variants such as Adaptive Multi-Layer Prioritized Fictitious Self-Play have appeared in 2023 to 2024 wargaming papers, and self-play is the backbone of beyond-visual-range air combat work and cyber attack-defense game solvers¹⁶.

2.5 Hybrid RL plus Symbolic, Scripted, or LLM Components

Pure RL is brittle, sample hungry, and hard to audit. The fastest-moving thread in 2024 to 2026 pairs RL with structured priors. Examples include the hierarchical hybrid combining DRL with scripted doctrine agents for wargaming⁵, LLM-guided hierarchical MARL for autonomous cyber defense¹⁵, and the U.S. Army CGSC exercise that adjudicated nine full turns with an LLM ontology layered over Monte Carlo dice instead of an RL policy⁶.

11. Black & Darken, Scaling AI for Digital Wargaming in Support of Decision-Making, NATO STO-MP-MSG-207, arXiv:2402.06075, arxiv.org/abs/2402.06075

12. DoD Data, Analytics, and AI Adoption Strategy 2023, [media.defense.gov \(DoD AI Adoption Strategy\)](https://media.defense.gov/DoD_AI_Adoption_Strategy)

13. Black & Darken, Scaling Intelligent Agents in Combat Simulations for Wargaming, I/ITSEC 2023, arXiv:2402.06694, arxiv.org/abs/2402.06694

14. Multi-Agent Cooperative Combat Simulation in Naval Battlefield, J. of System Simulation, 35(4), 2023, dc-china-simulation.researchcommons.org

15. Hierarchical Multi-Agent RL for Autonomous Cyber Defense in Coalition Networks, IEEE, [ieeexplore.ieee.org \(HRL air combat\)](https://ieeexplore.ieee.org/HRL%20air%20combat)

16. Hierarchical Reinforcement Learning for Air Combat at DARPA AlphaDogfight Trials, IEEE Aerospace, ieeexplore.ieee.org/document/9950612

3. Case Studies

3.1 DARPA AlphaDogfight Trials (2019 to 2020)

Sponsor	DARPA, managed by Johns Hopkins APL
Environment	F-16 flight simulator, within-visual-range air-to-air combat
Methods	Eight teams trained deep RL agents. Heron Systems used deep RL on a single policy. Lockheed Martin used hierarchical RL with maximum-entropy training and SME reward shaping. ⁴
Key result	Heron Systems defeated a USAF F-16 weapons instructor 5 to 0 in the final human-vs-AI match in August 2020. ¹
Follow-on	Heron was acquired by Shield AI in 2021. Winning techniques rolled into the Air Combat Evolution (ACE) program and flew on the X-62A VISTA in December 2022 and 2023. ^{2,17}
Significance	First widely-publicized superhuman defeat of an expert military pilot by RL. Triggered DoD investment in production-quality RL agents.

3.2 DARPA Air Combat Evolution (ACE) and X-62A VISTA Flights

Sponsor	DARPA (ACE) and AFRL (Autonomous Air Combat Operations), with Air Force Test Pilot School and AFTC
Environment	X-62A VISTA Gen2020 aircraft, simulating MQ-20, F-16, and other airframes in real flight at Edwards AFB
Methods	AI-driven autonomy agents trained in HPC simulation, transferred to live aircraft. Constructive AI red-team agents served as adversaries. ACE focused on WVR dogfighting; AACO on BVR and ISR. ²
Key result	12 live flight tests in December 2022 successfully executed dogfighting and BVR engagements. Agents were swapped on-aircraft in minutes. Tactical follow-on testing reported in 2023. ^{2,17}
Significance	First sustained sim-to-real RL transfer onto a manned fighter testbed. Demonstrated runtime safety trip system enabling rapid agent iteration on real hardware.

3.3 DARPA Gamebreaker (2020 onward)

Sponsor	DARPA, Defense Sciences Office
Environment	Existing open-world commercial wargames; later "Mosaic" multi-domain models
Methods	AI-driven analysis of game balance using algorithmic approaches inspired by AlphaGo and AlphaStar (referred to as "AlphaMosaic" equivalents). Goal is to derive a game-balance state equation and to find capability or rule modifications that destabilize the game. ¹⁸
Key result	Public program description states the goal of maximizing imbalance to create a military advantage, and seeking equilibrium when an adversary is doing the same. Specific quantitative results are not public. ¹⁸
Significance	Reframed wargaming AI from playing a fixed game to redesigning the game, that is, identifying which capabilities and rules most shift the win probability.

3.4 DARPA SCEPTER (2023 to present)

Sponsor	DARPA, Strategic Technology Office
Environment	Tailorable abstractions of trusted, expert-informed combat simulators; scenarios up to 10,000 agents
Methods	Unscripted, goal-oriented agents that explore the state-action space at 10,000 to 100,000 times real time; methods to linearize agent-agent interactions, enforce spatial locality, and produce a small set of human-interpretable courses of action (COAs). ^{3,19}
Key result	Program objectives include discovery and execution of distributed kill-chains, validation of the best COAs in high-fidelity trusted simulators, and quality assessment against quantifiable measures. ³
Significance	Most ambitious public DoD effort to apply machine-generated strategy directly to real warfare planning rather than to a tactical sub-problem.

3.5 Naval Postgraduate School Hierarchical RL for Combat Simulation

Sponsor	Naval Postgraduate School (Black and Darken), reported through NATO STO and I/ITSEC
Environment	Digital combat simulations representative of operational-level wargames
Methods	Hierarchical RL with multi-model decision-making, localized observation abstraction using piecewise-linear spatial decay, and dimension-invariant observation abstractions. Hybrid framework that uses RL for high-level decisions and scripted agents for low-level execution. ^{5,11,13}
Key result	Localized observation abstraction outperformed traditional global observations in both efficacy and compute efficiency. Hybrid AI framework showed better adaptability and reliability than non-hybrid baselines. ⁵
Significance	Identified concrete architectural tricks for scaling RL from squad to operational scenarios, a core open problem flagged by SCEPTER.

3.6 U.S. Army CGSC AI-Enabled Wargaming Exercise (Nov 2025)

Sponsor	U.S. Army Command and General Staff College, on the Army Vantage platform
Environment	Four-star joint task force operation in an Indo-Pacific theater against a peer A2/AD network; 32 student officers, including four coalition partners
Methods	Hybrid Pipelined Ontological-Augmented Generation (OAG) with Retrieval-Augmented Generation; custom Vantage LLM agent with a 128,000-token context, joint publications, and missile probability tables. Adjudication used Monte Carlo with hundreds of iterations per turn. RL was not used; LLMs and stochastic adjudication were used in its place. ⁶
Key result	AI-enabled teams completed 9 full turns in under 3 hours, versus 1 to 2 turns for analog teams in the same window, a 5x throughput gain. Each turn produced 100 to 200 doctrinally referenced data points and ran 15 to 25 minutes. ⁶
Significance	Demonstrates that non-RL AI (LLM plus Monte Carlo) can substantially accelerate staff-level wargames today, while RL approaches mature for higher-fidelity tactical problems.

3.7 DIU Thunderforge (2025) and USAF WarMatrix (2026)

Sponsor	Defense Innovation Unit with Scale AI (Thunderforge); Air Combat Command (WarMatrix)
Environment	Operational and theater-level planning at INDOPACOM, EUCOM, and the Joint Staff (Thunderforge); GE 26 Benchmark Wargame and Air Force force-design wargames (WarMatrix)
Methods	Commercial AI-powered modeling, simulation, and decision-support stacks. Architecture is LLM-centric for plan generation, with RL- or search-based exploration for COA development. Air Force RFI explicitly seeks simulations running 10,000 times faster than real time. ^{20,21,22}
Key result	Thunderforge prototype contract announced March 2025 with combatant command rollouts. WarMatrix went operational at the March 27, 2026 GE 26 Benchmark Wargame. ^{7,20,22}
Significance	Operational entry point for the SCEPTER-style "fast exploration" vision at theater scale; the integration target for academic RL work in the near term.

17. Shield AI buys Heron Systems, FedScoop, July 2021, [fedscoop.com \(Shield AI acquires Heron\)](https://fedscoop.com/shield-ai-acquires-heron/)
18. Gamebreaker program page, DARPA, darpa.mil/research/programs/gamebreaker
19. 3 ways DARPA aims to tame strategic chaos with AI, Breaking Defense, Sept 2023, [breakingdefense.com \(SCEPTER explainer\)](https://breakingdefense.com/scepter-explainer/)
20. Thunderforge Project announcement, DIU, March 2025, diu.mil/latest/diu-and-scale-ai-launch-thunderforge
21. US Air Force wants AI to power high-speed wargaming, Defense News, Dec 2025, [defensenews.com \(Air Force AI wargaming RFI\)](https://defensenews.com/air-force-ai-wargaming-rfi/)
22. USAF GE 26 showcases new AI-enabled WarMatrix wargame system, Air Combat Command, April 2026, [acc.af.mil \(Article 4150428\)](https://acc.af.mil/article/4150428)

4. Comparison Table: Methods, Domains, and Reported Results

Where multiple results are reported, the most operationally meaningful is shown. Where the source did not publish quantitative results, that is noted explicitly.

System / Study	Method	Domain	Reported result	Limitation
Heron Systems, AlphaDogfight 2020	Deep RL, self-play	F-16 sim dogfight (WVR)	5 to 0 vs. USAF weapons instructor ¹	Single airframe, no sensors-of-opportunity, brittle to scenario shifts
Lockheed Martin (Pope et al. 2021)	Hierarchical RL with maximum-entropy and reward shaping	AlphaDogfight, sim air combat	2nd of 8, defeated USAF F-16 weapons instructor in match play ⁴	Requires expert reward design; hierarchy hand-tuned
DARPA ACE on X-62A (AFRL 2022 to 2023)	Sim-trained deep RL, sim-to-real transfer	Live F-16-class flight, WVR and BVR	12 live flights; in-flight agent swapping; tactical testing 2023 ^{2,17}	Sim-to-real gap; safety envelope still requires human override system
DARPA Gamebreaker	AlphaGo / AlphaStar-style search and RL over commercial games and Mosaic models	Multi-domain force composition	Program targets explicit balance-equation model; quantitative results not public ¹⁸	Transfer from commercial games to actual force design is unproven
DARPA SCEPTER	Unscripted goal-oriented agents, scalable abstractions, COA validation in trusted sims	Theater-level COA generation, up to 10,000 agents	State-space exploration 10,000 to 100,000 times real time ³	Open question whether discovered COAs survive high-fidelity validation
Black & Darken (NPS) 2023 to 2024	Hierarchical RL plus scripted hybrid, localized and dimension-invariant observations	Wargaming combat simulation	Improved efficacy and compute efficiency vs. global-observation baselines ^{5,11,13}	Reported in academic and NATO venues; not yet a fielded system
MARL-CCSA naval cooperative combat (2022)	MARL with attention and decoupling-priority replay	Naval surface battle simulation	Faster convergence and higher win rate than baseline MARL ¹⁴	Custom env, no operational integration claimed
Dabholkar et al. (JDMS 2024)	Adversarial attacks on A3C and PPO agents	StarCraft II C2 maps	Demonstrated severe performance degradation under environment perturbations ⁸	Establishes need for robust training; mitigations not yet quantified at scale
Lamparth et al. (Stanford 2024)	LLM agents in wargame role-play, compared to 214 human national-security experts	U.S.-China crisis escalation wargame	LLM agents more aggressive than humans and highly sensitive to scenario wording ¹⁰	Behavioral fidelity to expert humans is limited; should not yet be used to replace humans

System / Study	Method	Domain	Reported result	Limitation
U.S. Army CGSC AI-Enabled Wargame (2025)	LLM Ontology-Augmented Generation plus Monte Carlo adjudication (no RL)	JTF-level Indo-Pacific COA wargame	9 turns in under 3 hours vs. 1 to 2 analog turns, 5x throughput ⁶	Adjudication realism depends on doctrine prompts; not a closed-loop learning system
DIU Thunderforge (Scale AI, 2025)	Commercial LLM stack with search and simulation for theater planning	INDOPACOM and EUCOM operational planning	Prototype contract awarded; rollout to combatant commands announced ²⁰	Outcome metrics still pre-operational; commander trust calibration in progress
USAF WarMatrix (2026)	Cloud-based AI wargame hub targeting 10,000x real-time simulation	Air Force force design and benchmark wargames	Operational debut at GE 26 Benchmark Wargame on 27 March 2026 ^{7,21,22}	Public metrics still limited; clearance-bounded reporting

5. Success Metrics in the Literature

Reported metrics fall into four buckets. Few papers report all of them; the most defensible studies report at least metrics 1 and 4.

Metric	What it measures	Representative use
Win rate / score	Fraction of engagements won against a fixed or human baseline.	AlphaDogfight 5-0 vs human; MARL-CCSA naval combat ^{1,14}
Elo and league rank	Skill in a population of opponents during self-play.	Fictitious self-play wargame variants, beyond-visual-range air combat ¹⁶
Sample efficiency	Reward attained per environment step or wall-clock hour of training.	Inverse-variance RL (ICLR 2022) for noisy supervision; NPS observation abstractions ^{5,23}
COA quality and novelty	Whether discovered courses of action are operationally relevant, surprising, and interpretable.	DARPA SCEPTER quality assessment against quantifiable measures ³
Throughput and wall-clock	Decisions or turns per hour, faster-than-real-time factor.	CGSC AI wargame 5x throughput; WarMatrix 10,000x real-time target ^{6,21}
Robustness	Performance retained under adversarial perturbations or distribution shift.	Dabholkar et al. adversarial attacks on PPO and A3C C2 agents ⁸
Human-alignment fidelity	How closely agent behavior matches expert humans on the same scenario.	Stanford LLM vs 214 experts crisis-escalation wargame ¹⁰

6. Persistent Challenges

- Sample efficiency. Tactical agents typically need tens of millions of simulator steps. Inverse-variance RL and uncertainty-weighted updates improve efficiency on noisy supervision²³, but theater-scale problems remain compute bound. SCEPTER is explicitly attacking this through model abstraction.³
- Sim-to-real transfer. ACE flights on the X-62A succeeded in part because of a runtime safety-trip system that allowed rapid in-flight agent updates; the underlying sim-to-real delta remains a primary risk.²
- Reward hacking and gaming. Self-play in open-world games (Gamebreaker) exposes agents that exploit simulator artifacts. This is the source of the program's pivot toward studying balance itself as the object of learning.¹⁸
- Adversarial brittleness. Dabholkar et al. demonstrate that PPO and A3C C2 agents on StarCraft II maps degrade sharply under noise injection and environment perturbations, recommending more robust training algorithms for battlefield deployment.⁸
- Explainability and commander trust. Both SCEPTER and Air Force adaptive-wargaming commentary call out the need for a small number of human-interpretable COAs. Explanation of multi-agent RL aerial combat decisions is an active research thread.^{3,24}
- Behavioral fidelity to humans. LLM wargame agents have shown systematic aggression bias and high prompt sensitivity, which makes them unsafe substitutes for human red teams without correction layers.¹⁰

- Classification and data access. The largest sim-to-real datasets and most realistic OPLAN scenarios are classified, which biases public research toward proxy environments such as StarCraft II, custom F-16 sims, and synthetic naval and ground engagements.

23. Mai et al., Sample Efficient Deep RL via Uncertainty Estimation, ICLR 2022, arXiv:2201.01666, arxiv.org/abs/2201.01666

24. Explaining Strategic Decisions in Multi-Agent Reinforcement Learning for Aerial Combat Tactics, recent IEEE / journal literature on explainable MARL.

7. Comparison: RL vs. Other AI Techniques in Wargaming

Technique	Strengths in wargaming	Weaknesses
Classical search (MCTS, alpha-beta) and behavior trees	Auditable, no training data needed, strong on bounded tactical problems with known rules.	Exponential blow-up with agent count and state dimension; struggles in continuous control and partial observability.
Scripted and rule-based agents	Doctrine-consistent, explainable to commanders, fast to certify for use in training simulators.	Predictable; cannot generate novel COAs; force-on-force training becomes scripted target practice.
Supervised imitation learning	Captures expert human style; useful as a warm start.	Expert traces are rare and classified; agents cannot exceed demonstration quality.
Deep RL (PPO, SAC)	Discovers novel tactics, scales to continuous control, no labeled data required.	Sample inefficient, brittle to perturbation, hard to explain.
Hierarchical and MARL	Maps to command echelons; supports coordinated forces and decentralized execution.	Credit assignment, non-stationarity, and stable training remain open problems.
Self-play / population-based training	Generates adaptive red teams; produces emergent novelty.	Can converge to non-doctrinal local optima; expensive compute.
LLM-based planning and adjudication	Excellent for staff-product generation, doctrine reasoning, and rapid turn adjudication.	Cannot learn from outcomes without an external loop; aggression and prompt-sensitivity biases.
Hybrid RL + LLM + scripted	Combines doctrine, narrative reasoning, and adaptive policy. Pattern adopted by SCEPTER, NPS HRL work, and CGSC pilots.	Integration complexity; harder to certify; partition of responsibility between components is not yet standardized.

8. Future Outlook for DoD Wargaming (2026 to 2030)

8.1 Hybrid LLM plus RL planners become the default

Thunderforge and WarMatrix point to an architecture where an LLM reasons about doctrine, orders, and narrative, while an RL or search layer explores COAs in a trusted simulator. Expect the next two years of academic work to formalize the LLM-to-RL interface and develop standard evaluation benchmarks analogous to AlphaDogfight for staff-level wargaming.^{20,22}

8.2 Faster-than-real-time simulators unlock theater RL

The Air Force 10,000x real-time requirement, combined with SCEPTER's 10,000 to 100,000x target, implies a sustained investment in differentiable or GPU-resident combat simulators. This is the single largest lever for sample efficiency at theater scale.^{3,21}

8.3 Adversarial robustness becomes a procurement requirement

Published vulnerabilities of PPO and A3C in C2 settings, combined with operational use in Thunderforge and WarMatrix, will pull adversarial robustness from academic concern to procurement checklist. Expect red-team and certification frameworks modeled on cyber Authority to Operate processes.⁸

8.4 Explainability and human-machine teaming take center stage

SCEPTER explicitly limits output to a small number of human-interpretable COAs. Expect interpretable surrogate policies, counterfactual reasoning over agent decisions, and language-grounded explanations to become standard deliverables alongside the policy itself.^{3,24}

8.5 Sim-to-real and live-virtual-constructive integration

ACE on the X-62A is a proof that RL agents can fly real fighters within a safety envelope. The next step is integration into live-virtual-constructive exercises, where RL agents act as constructive opponents (or partners) alongside live forces during major joint events.^{2,17}

8.6 Trust calibration, ethics, and behavioral alignment

The Stanford LLM-vs-human wargame study is the cleanest public evidence that off-the-shelf LLMs do not yet mirror expert human decision-making in crisis. The same is true, by extension, for any RL or hybrid agent trained without behavioral alignment to doctrine and rules of engagement. Expect alignment evaluations to become a gating criterion for use in real decision support.¹⁰

8.7 Recommendations for DoD-relevant programs

- Invest in shared, fast simulators. Compute-efficient, multi-domain simulators with standard APIs (analogous to OpenAI Gym for tactical combat) would unlock the most leverage.
- Standardize an RL-LLM interface. Treat the LLM as the planner and the RL agent as the trusted explorer; standardize their handoff so components can be swapped per scenario.
- Require adversarial robustness benchmarks. Mandate evaluation against published attack suites before any RL agent enters operational wargaming.
- Run continuous human-in-the-loop wargames. Pair every fielded agent with structured human red-teaming at CGSC, NWC, and Air University to catch reward hacks early.
- Publish at least adjudication-level results. Even where policies must stay classified, publishing throughput, novelty, and robustness metrics would accelerate the academic-to-DoD feedback loop.

9. References

All sources are publicly available. Reference numbers correspond to the superscript citations used throughout the report.

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3. SCEPTER program page, DARPA, <https://www.darpa.mil/research/programs/scepter-strategic-chaos>
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